FLARE: A Framework to Reduce Hallucination in Generative AI

S Srushti Kashid Follow 8 min read · 5 days ago

-By Srushti Kashid

Introduction

The AI based on neural networks has achieved significant advances in natural language processing. GPT-4, PaLM, and LLaMA models can currently perform multi-step reasoning and provide human-like answers in a variety of fields. Their use in areas like education, healthcare, and customer support shows how large language models (LLMs) are becoming part of everyday systems.

Hallucination remains a significant problem, despite improvement. LLMs tend to provide assured responses, which are incorrect or lack support by any factual data. This impacts on their credibility particularly in areas where factual accuracy is of essence. In order to minimize this issue, the scholars proposed Retrieval-Augmented Generation (RAG) that enables models to use external sources during text generation.

However, the vast majority of RAG systems only access information once, prior to generation. This proves to be a constraint in cases that are lengthy or have several steps since the model is tied to the set context.

FLARE (Forward-Looking Active Retrieval-Augmented Generation) gives this a better shot by turning retrieval dynamic. It allows the model to make predictions on what it requires next, identify uncertainty, and fetch new information during generation.

The paper is divided into the following sections: Section 2 outlines the background, Section 3 outlines the FLARE framework, and the conclusion is presented in Section 4.

Background

2.1 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is a current methodology built on top of retrieval with the generation ability of big language models (LLMs). RAG enables the system to search external databases such as Wikipedia, databases, or online repositories when generating text in addition to depending entirely on the pre-trained memory that the model

This process enhances the factual accuracy and allows models to have access to the current information that could not be available in the training information. RAG has demonstrated good performance in its activities such as answering questions and summarization whose output is founded on confirmed documentation. Nevertheless, there exist most RAG systems that do not revisit retrieval again after the generation sets in. The retrieval is a weakness in the longer or multi-step tasks because the context might change or the initial information can be outdated. Consequently, the model can be hallucinated by the absence of details or the staleness of context.

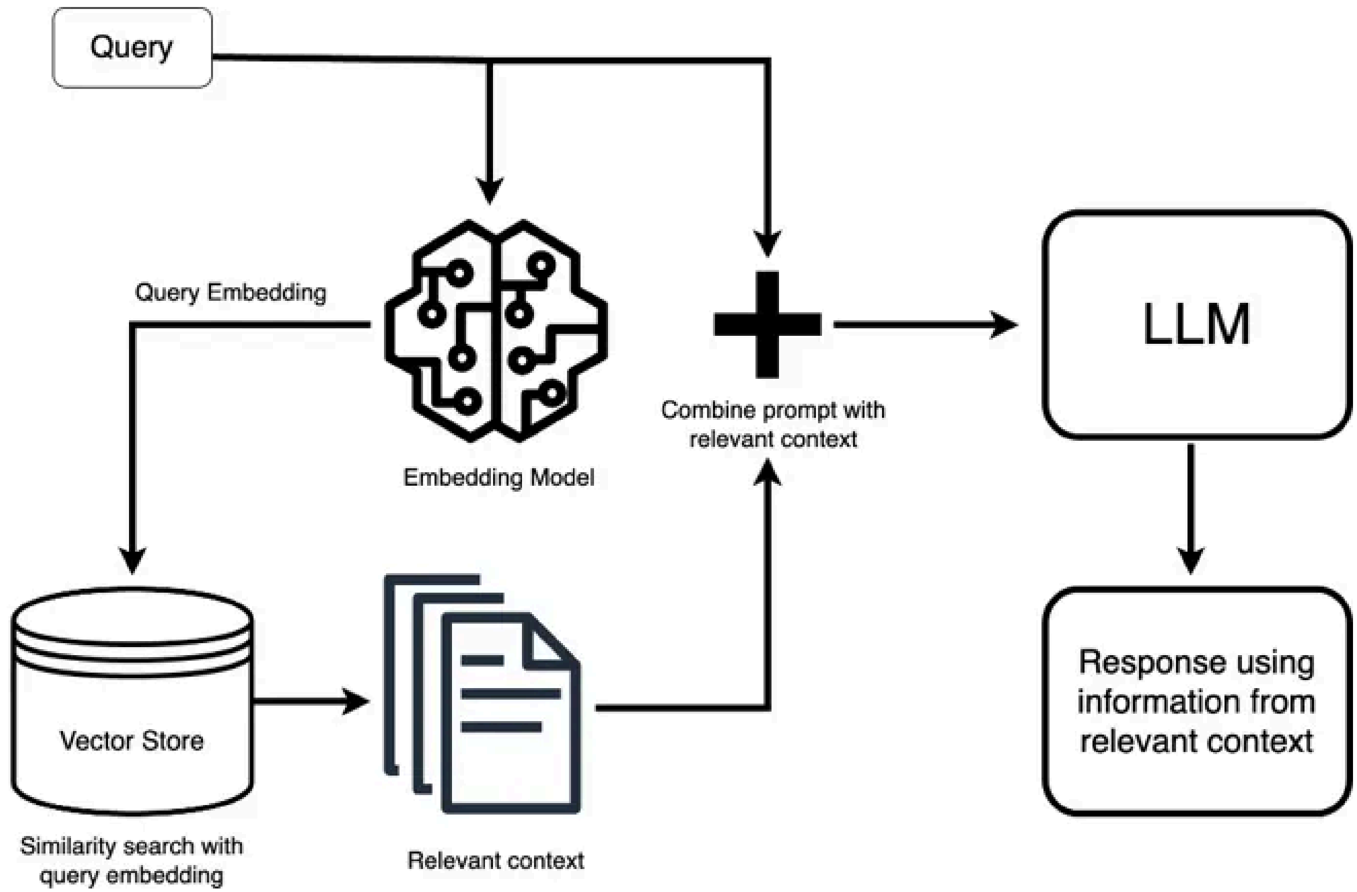

Figure 1 Workflow of a Retrieval-Augmented Generation (RAG) system.

A standard RAG pipeline contains the following components as shown in Figure 1:

· Input/ query: The input or question of a user that starts the process.

· Embedding Model: The model turns the query/input into a numerical representation (vectors) of the semantic meaning of that query.

· Vector Store: The output from embedding model — vectors are stored in vector database and also similarity searches are used to retrieve the relevant documents or passages.

· Relevant Context: The information that is retrieved from a vector database.

· LLM (Large Language Model): It is used to generate a response based on combining the query and retrieved context so that the generated outputs are more accurate and grounded.

RAG systems have the advantage of being more reliable and situation-specific in their response because they utilize external knowledge in the creation of the response instead of relying on internal memory alone.

2.2 Hallucination: Causes and Risks

Hallucination happens when language model produces text that appears to be accurate but not verifiable by facts or other sources. This is due to the fact that the model is based on the trends observed through the learning process and not through real-time validation. In cases where the model has inadequate information regarding some topic, it has a tendency of filling the gaps in order to stay fluent, consequently producing confident yet possibly wrong outputs.

Biased training data, outdated knowledge or limited access to verified sources usually contribute to hallucinations of LLMs. The practice of introducing false representations in the academic or professional sense undermines the trust in AI-obtained results and may spread misinformation in the decision-making process that requires caution. Hallucinations may cause severe misinterpretations and even moral or legal implications in such serious areas of life as healthcare, finance, and law. In every application, inaccurate answers cause a lack of confidence in users and deter further use of AI systems.

The FLARE Framework

FLARE (Forward-Looking Active Retrieval-Augmented Generation) is based on retrieval of result in a dynamic sense. The section defines the working process of FLARE in four areas, which include (3.1) forward-looking prediction, (3.2) uncertainty detection, (3.3) query formulation and interleaved generation, and (3.4) efficiency mechanisms.

3.1 Forward-Looking Prediction

Forward-looking prediction is the main concept of FLARE. Whereas the traditional systems activate retrieval through the advancement of tokens, FLARE is able to anticipate what to write next and retrieve information in advance.

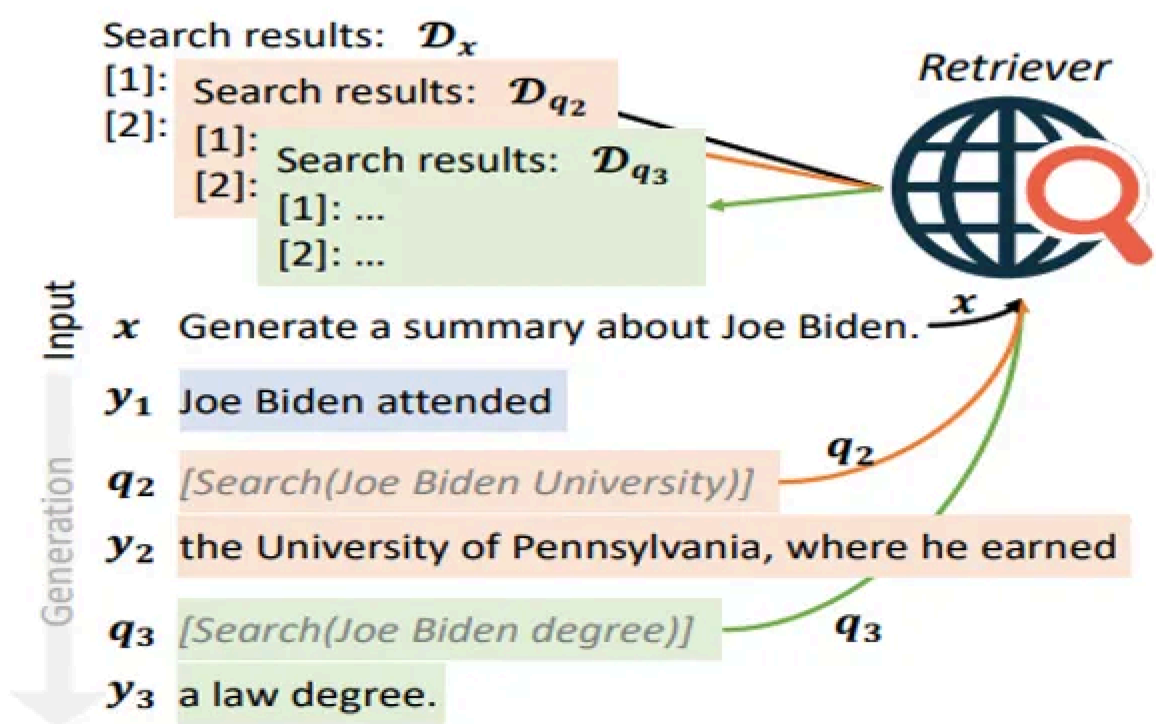

Figure 2: An illustration of forward-looking active retrieval augmented generation with retrieval instructions

Figure 2 shows the working of FLARE framework that is how FLARE is used to retrieve information in a dynamic way. The process starts as shown in

Figure 2 when the user enters a task prompt i.e. **Generate a summary about Joe Biden**. The model begins to produce a first word phrase like, Joe Biden attended. At this stage, the model understands that the sentence needs some information to proceed properly like the name of the university. It automatically gives the structured query such as [Search (Joe Biden University)] and response is generated.

The model predicts content, and if there is absence of some detail (his university) and the model returns a structured query such as [Search (Joe Biden University)]. Once it retrieved the information it required, FLARE can complete a full and correct sentence: "Joe Biden went to the University of Pennsylvania…". FLARE predicts the next required information by making use of the predict–retrieve–generate cycle, enabling the model to anticipate informational needs in advance. This approach is effective for long-form text generation because it makes the response accurate, consistent, and coherent.

3.2 Uncertainty Detection and Triggering Retrieval

FLARE relies on uncertainty detection in order to determine when it is to be retrieved. The model at every generation stage compares the confidence level of every predicted token. When all the tokens are above certain threshold, it will keep on writing. However, when any of the tokens drops below that threshold, the retrieval is automatically activated. Upon activation, FLARE will create a query on the low-confidence phrase and retrieve the relevant documents, which are then added to the context and proceeds further.

Mathematically, above dynamic retrieval process can be summarized as:

$$y_t = \begin{cases} \hat{s}_t, & \text{if all token probabilities} \geq \theta \\ LM([D_{q_t}, x, y_{<t}]), & \text{otherwise} \end{cases}$$

In the mathematical representation given above, the model will only continue to write when it is sure of its predictions. In case of uncertainty, it recalls the new evidence and paraphrases the sentence with the new evidence. This self-regulated mechanism allows FLARE to base every statement on proven knowledge and achieve efficiency and accuracy at the same time.

3.3 Query Formulation and Interleaved Generation

After identifying uncertainty, FLARE needs to make decisions on what to search and utilization of the results. This step — query formulation is very important since the quality of the data retrieved directly influences the final result. FLARE uses two strategies to generate retrieval queries:

I. Implicit masking

II. Explicit question generation

1. Implicit Query Formulation

When the model identifies low-confidence tokens within a predicted sentence, those uncertain words are masked before retrieval. As an example, when the temporary sentence is: Joe Biden attended [MASK], where he earned [MASK], the masked tokens indicate areas where the model lacks confidence. These masked words are sent as a query to the retriever which tries to locate the information that is missing in the external sources like Wikipedia or a web index. FLARE eliminates the possibility of error contamination by avoiding the use of tokens that might be erroneous and only the questionable content drives the search.

2. Explicit Query Formulation

In Explicit Query Formulation low confidence texts are converted to human-readable questions. As an example, the uncertain part above would create queries like:

What was the university Joe Biden went to?

What is the level of education that Joe Biden received?

These are explicit questions that help the system understand exactly what information is missing, making the retrieval process more precise and reliable. As a result, the system can provide high quality, context specific documents. This method is similar to the way, in which human beings define the unknown through specific questions.

3.4 Interleaved Generation and Retrieval

Once the query has been formulated (implicitly or explicitly), the retriever provides a ranked list of documents relevant to the query. The FLARE then combines retrieval and generation: the already retrieved evidence is prepended to the current context and the incomplete sentence is regenerated by the model with the help of the new information. This process maintains itself until the next uncertainty is met whereby the model proceeds to the next stage.

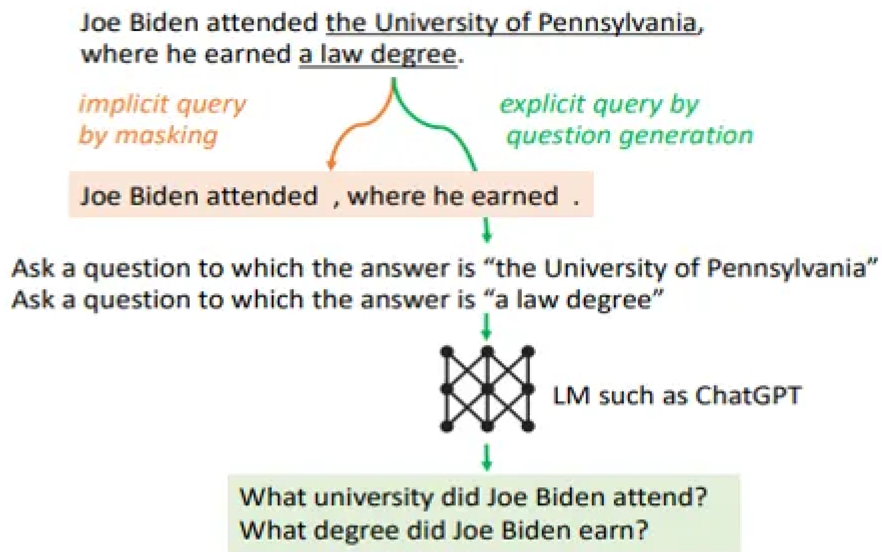

Figure 3: Implicit and explicit query formulation.

This back-and-forth writing, detection, retrieval and regenerating style ensures that the output of the model is consistent and well-grounded throughout. This mechanism is shown in figure 3 below. It demonstrates how FLARE discovers uncertain tokens, masks or converts them into questions, retrieves supporting information and puts the findings back into the generation stream to ensure the accuracy and coherence of facts.

Efficiency Factors

However, despite the fact that FLARE enhances factual accuracy by means of active retrieval, it also needs to be efficient. Constant retrieval may increase input size and computation, thus FLARE employs 3 optimization strategies, namely, confidence thresholds, masking thresholds, and context pruning, to trade off accuracy and performance.

i) Confidence Threshold (θ):

This is a threshold at which retrieval is initiated. The increasing values enhance reliability at the cost of processing time whereas the decreasing values are fast at the expense of not updating. A value of 0.6 to 0.8 gives the best balance.

ii) Masking Threshold (β):

This regulates the numbers of low-confidence tokens that are substituted when forming a query. Greater β values generalize queries and less β enhances specificity. A value of 0.3 to 0.5 produces consistent performance at work.

iii) Context Pruning:

With continued generation, the old information would be eliminated to create space. FLARE is coherent because it keeps only recent and relevant information, which does not overload the system.

Combining these mechanisms, FLARE will only retrieve when it is required and rewrites with uncertain segments efficiently and text generation is accurate and scalable.

Conclusion

The Forward-Looking Active Retrieval-Augmented Generation (FLARE) model provides an effective and flexible approach to one of the most lasting issues of generative AI — hallucination. FLARE incorporates forward-looking prediction, uncertainty detection, intelligent query formulation and context management converting retrieval into a dynamic system with real time feedback. This enables large language models to predict their own knowledge requirements, find useful evidence during text generation, and base their answers on knowledge that has been proven true.

In general, FLARE enhances the reliability and the efficiency of language models in terms of factual accuracy, and it is particularly useful in long-form and knowledge-based assignments. It is a modular design that can be easily implemented with the pre-existing AI systems so that organizations can launch more transparent, accurate, and trustworthy generative solutions. As studies progress, the paradigm of AI systems such as FLARE will be instrumental in determining a future where information generated by AI systems is not only fluent and coherent, but also verifiably true.

+ Rag + Hallucinations + Llm Hallucinations LLM + Vector Database

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Written by Srushti Kashid

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